The Two Layers – Ratios and Conversion.

Johann Pascher

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Preface

This document is for readers who want to enter the methodological architecture of FFGFT without first working through its full mathematical apparatus. It does not repeat any derivation; it explains a reading order.

The content is not new. The central claim — that FFGFT operates on two cleanly separated layers — already appears at various points in the document series (above all in Docs. 014, 015, 044, and 134). But there it stands as a technical clarification alongside other technical remarks, not as an architectural principle with its own voice. That is what this document tries to provide.

A first-time reader otherwise risks getting stuck at a point that resolves itself the moment one recognizes the two layers. Whoever does not recognize them sees in every fractal correction factor a free parameter and in every natural constant an independent quantity — missing the actual point: that FFGFT operates with exactly one free parameter $\xi_0 = 4/30000$, and that every other quantity is either a pure ratio or arises through a geometrically determined scale-bridge.

.1 The starting point: the constants problem

Ordinary school physics presents a list of "fundamental constants of nature":

- the speed of light c,
- the reduced Planck constant $\hbar$,
- the gravitational constant G,
- the Boltzmann constant k_B,
- the electric constant ε_0,
- the fine-structure constant $\alpha \approx 1/137$,
- the elementary charge e,

and so on. Each has its own unit, its own value, each appears independent. The student sooner or later asks: *Why precisely these values? Why not others?*

The usual answer, slightly abbreviated, runs: "We don't know. That's just how it is."

This answer is intellectually unsatisfying. It suggests that nature is equipped with a dozen independently adjustable knobs, and that humans are to measure them off without ever being able to understand them. Sommerfeld, Eddington, Dirac, and Feynman never accepted this; each in his own way tried to see behind the wall of "fundamental constants".

FFGFT makes the same attempt, but on a different path. The path begins with a seemingly harmless observation: many of these "constants" are not really constants at all. They are *exchange rates between choices of scale*, not independent natural quantities. This becomes visible only when one reduces consistently, in two steps.

.2 First step: $\hbar = c = G = k_B = 1$

The first step is standard and has been common in theoretical physics for decades: one works in *natural units*, in which the constants $\hbar$, c, G, and k_B are simply set equal to one (Docs. 014, 015).

What at first appears to be notational convenience is in fact a structural statement. In SI units, length (meter), time (second), and mass (kilogram) have independent dimensions. This independence is, however, a *convention*, not a property of nature. Setting $c = 1$ makes length and time the same quantity. Setting $\hbar = 1$ makes energy and inverse time the same quantity. Setting also $G = 1$ and $k_B = 1$, *all* mechanical, thermodynamic, and gravitational quantities collapse onto a single dimension: energy.

Quantity	Dimension
Energy, mass, temperature, frequency, momentum	$[E]$
Length, time, wavelength	$[E]^{-1}$
Force	$[E]^2$
Velocity	dimensionless
Electric charge	dimensionless

This table (more fully in Doc. 015) says something simple but important: *the apparent variety of physical quantities is to a large extent an artifact of the choice of scale*. There is only one fundamental dimension, and all others are powers of it. The "many constants" of school physics are no longer independent — they are conversion factors between different energy powers.

This step alone is not specific to FFGFT. It belongs to the standard toolkit of high-energy physics. But it is the precondition for the second, less familiar step.

.3 Second step: $\alpha = 1$

Even in natural units with $\hbar = c = G = k_B = 1$, one quantity seems to remain fundamental, unexplained, "magical": the fine-structure constant

$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c} \approx \frac{1}{137{,}036}.$$

Feynman called it “the greatest mystery of physics”: a dimensionless number that appears out of nowhere and yet governs the whole of chemistry and atomic physics.

Here FFGFT takes a decisive second step: *α may also be set equal to one, in a consistently extended system of units*. This is not a trick, not convention in the sense of mere convenience, but a structural insight known in physics under the name **Heaviside-Lorentz convention** and discussed at length in Doc. 044:

- In SI units: $\alpha = e^2/(4\pi\varepsilon_0\hbar c) \approx 1/137$.
- In Gaussian units: $\alpha = e^2/(\hbar c) \approx 1/137$.
- In Heaviside-Lorentz units (with additionally $4\pi\varepsilon_0 = 1$ and a suitable definition of charge), $\alpha = e^2/(4\pi)$, and in a further-extended system even $\alpha = 1$.

All three systems describe the same physics. The numerical value $1/137{,}036$ is therefore *not* a property of the world, but a property of how we parametrize the world. What changes is only *where* the information sits — in α, in the charge e, or in the electric constant ε_0.

Only after one has seen through this does the real point become visible: *exactly one dimensionless quantity remains that cannot be removed by any choice of scale*. In FFGFT this quantity is the geometric parameter

$$\xi_0 = \frac{4}{30000} \approx 1.33 \times 10^{-4}.$$

Everything else — c, $\hbar$, G, k_B, α, and even particle masses, insofar as they are taken as pure ratios of one another — follows from ξ_0 alone, or is a scale convention.

Important clarification: “set to 1” means “factored out”

The values $c, \hbar, G, k_B, \alpha$ that have been “set to one” do not disappear from physics. They are *temporarily factored-out scale factors*, not quantities rationalized away.

As long as one reckons with *ratios* — mass ratios, frequency ratios, dimensionless couplings —, these factors cancel out, and no numerical substitution is needed. The answer is sharp without c or $\hbar$ ever appearing.

But the moment one wants to generate a *projected measurement value* in SI units — the electron mass in kilograms, a frequency in hertz, an energy in joules —, the values that were set to one must be *re-inserted*, together with the fractal correction. They are therefore conditional bridge factors: absent in the pure geometry, present in the projection. This is not a contradiction but the clean form of the architecture: whoever stays inside the geometry does not need them; whoever wants to see a measurement value brings them back.

.4 The insight: two layers

From these two steps follows the methodological architecture that gives this document its name. FFGFT operates on *two cleanly separated layers*:

Layer 1 — the ratios. Everything that can be expressed as a pure ratio between two quantities of the same kind — mass ratios such as m_μ/m_e, frequency ratios, winding numbers on the torus, dimensionless couplings, the parameter ξ_0 itself — is *unit-invariant*. These ratios are what is "really there" independent of any choice of scale. FFGFT makes its fundamental statements on this layer.

Layer 2 — the conversion to SI. The moment one asks how many kilograms a certain mass has, how many seconds a certain duration lasts, or how many volts a certain potential carries, one enters a second layer: the translation between the dimensionless ratios of Layer 1 and the humanly chosen units kilogram, second, meter, volt. This translation requires a *scale-bridge*, and in this bridge — and only in this bridge — the fractal corrections sit.

This is the central, often overlooked statement:

> **Fractal geometry sits in the ratios themselves — but there as a closed, frozen recursion.**
>
> **It becomes visibly active only in the bridge to the SI language, where the embedding correction must be made explicit.**

The ratios of Layer 1 are not fractal-free — they *are* fixed points of fractal recursions that have come to rest. What is absent is not the fractality but its visible motion. Only when one asks *how many kilograms* a mass has does the fixed point break open, and the embedding correction must be carried along explicitly. This transition — not the physics itself, but the representation of physics in human units — pays the "price" that Doc. 185 (*Time as the embedding cost of mathematics*) speaks of.

.5 Where the recursion is hidden

At this point a misunderstanding can arise. If Layer 1 contains the pure ratios, sharp and without correction — where then is the recursion of which FFGFT, as a *fractal-geometric* theory, speaks? Has one lost it among the ratios?

The answer is: no. It is preserved in the ratios. But there it looks still.

A plain number like the ratio $m_\mu/m_e \approx 206.768$ looks static. But it *is* the endpoint of a recursion that has come to rest in this number. Just as the golden section $\varphi = (1+\sqrt{5})/2$ apparently is a simple number but in truth the limit of the Fibonacci sequence $a_{n+1} = a_n + a_{n-1}$ — the recursion does not sit beside φ, it *is* φ. Just as π sits in the n-gons and e in the compound-interest formula: a pure number is always a frozen process. A *fixed point* is a geometry that has absorbed its own motion.

The FFGFT ratios are no different. The lepton ratio m_μ/m_e is the fixed point of a winding recursion on the 4D-torus, which converges after three generations with step size $\xi^{1/3}$. The rational prefactors $4/3, 16/5, 25/9$ (Doc. 205) are not arbitrary numbers but standing waves of a recursive mode structure. The deeper mechanism behind the choice of just these values is a Fibonacci-like convergence: the recursion amplitude $\varphi^{-2n}/\sqrt{5}$ reaches the order of ξ_0 at $n \approx 8$, and exactly there the mode sequence closes into a finite table (Docs. 011, 159). Three generations of step size $\xi^{1/3}$, terminated at Fibonacci convergence depth $n \approx 8$ — this is not coincidence but geometric necessity that only eight levels deep *can* be reckoned before the convergence amplitude drops below ξ.

The recursion is thus fully present in the ratios — but it has been put to rest because it has converged. What we see as a "ratio" is the final form of a motion that has come to rest. Whoever sees a ratio sees a finished story. Whoever asks *why exactly this* ratio, asks after the recursion that led to this fixed point. FFGFT answers on both levels: the ratios are the results, the windings are the underlying story.

.6 Where the recursion becomes visible

If the ratios are closed in themselves — where then does the recursion become visible *in motion*? Answer: in the bridge to the SI language.

An SI unit (kilogram, second, meter) is not a fixed point; it is an *arbitrarily chosen intermediate station* on the scale ladder of nature. The Planck scale and the electroweak scale lie about 17 orders of magnitude apart; the kilogram sits somewhere in the middle, fixed by a historically grown choice of scale, not by a geometric eigenvalue. Whoever translates from a geometric eigenvalue in Layer 1 to a kilogram value in Layer 2 does not fall from one finished recursion into another finished one, but moves along a scale staircase that is at this moment still *in motion* — not yet converged.

It is exactly there that $K_{\text{frak}} = 1 - 100\xi$ appears. The factor 100 is not ad hoc but the cumulative effect of the minimal deviation ξ over all rungs between Planck and electroweak scale. The logarithmic distance $\ln(10^{17}) \approx 39$, geometrically averaged over the relevant rungs, yields the amplification factor ~ 100. It is a recursion that has *not yet come to rest*, because our choice of scale cuts it off in the middle (Doc. 133).

The fractal geometry thus shows itself in two ways:

- in Layer 1 as *recursion at rest* — in the ratios that are fixed points of a converged winding sequence;
- in the bridge as *recursion in motion* — in K_{frak}, the cumulative trace of a scale cascade that we interrupt mid-way by our choice of units.

Both are the same geometry. Once as fixed point, once as scale flow. What looks like "two different phenomena" is the same fractal self-similarity in two states: at the goal and on the way.

.7 Consequences

From this architecture four consequences follow that noticeably simplify the understanding of many individual FFGFT results.

Ratio predictions are sharp

When FFGFT makes a statement about a *ratio* — for instance about m_μ/m_e or about the muon-to-tauon lifetime ratio (Doc. 160) —, this prediction is sharp: *without* fractal correction, because the correction cancels in a ratio of quantities of the same kind.

> When both quantities in numerator and denominator pass to the SI scale through the same conversion factor, this factor cancels out. What remains is pure geometry.

For this reason FFGFT ratio predictions are typically accurate to four or five decimal places. They test *the geometry*, not the conversion.

SI values carry the fractal signature

The moment FFGFT makes a statement about an *absolute SI value* — for instance about the electron mass in kilograms, or about the gravitational constant in m^3/(kg·s^2) —, the fractal correction appears. This is not a deficit of the theory but a necessary consequence of the architecture:

- In Layer 1 the electron mass is the geometric number $m_e^{\text{T0}} = 0.511$ (Doc. 014).
- The conversion to kilograms requires the scale factor $S_{T0} \approx 1.782662 \times 10^{-30}$, which is not chosen arbitrarily but derived from the fractal geometry.
- Only the combination of both — geometric number times scale factor — yields the experimental value $m_e^{\text{SI}} \approx 9.1094 \times 10^{-31}$ kg.

The fractal correction $K_{\text{frak}} = 1 - 100\xi \approx 0.9867$ (Doc. 133) belongs in this bridge, not in the electron mass itself.

D_f is a property of the bridge

A clarification on the fractal dimension $D_f \approx 2.999867$ follows: it does not sit "in the mass", nor "in space" in the naive sense, but in the bridge between geometrically natural language and SI language.

- The *local* fractal dimension of spacetime $D_f^{\text{space}} = 3 - \xi \approx 2.9999$ is a property of the vacuum and supplies only corrections of order 10^{-5}, that is, practically one.
- The *effective* fractal dimension $D_f^{\text{eff}} \approx 2.973$ that appears in the bridge is a *cumulative* quantity: it arises from the effect of the local deviation ξ over the full renormalization-group flow from the Planck scale to the electroweak scale (Doc. 133).

Both have the same geometric origin; but they sit at different positions of the architecture, and they must not be confused.

K_{frak} is not a fitting constant

A common question to FFGFT runs: "Is $K_{\text{frak}} = 1 - 100\xi$ not simply chosen so that the experimental values come out?" The answer becomes clear through the two-layer architecture:

- K_{frak} is *not* a free constant. It is the only geometrically possible bridge function between Layer 1 and Layer 2, given by the cumulative effect of ξ over the RG flow.
- The factor 100 is not ad hoc but follows from the logarithmic distance between Planck and electroweak scale ($\ln(10^{17}) \approx 39$) and the geometric averaging over the relevant rungs (Doc. 133).
- Once $\xi = 4/30000$ is fixed, K_{frak} is *compulsorily* determined.

There is therefore still only one free parameter: ξ_0. Everything that appears in Layer 2 follows from this one parameter and from the geometry of the scale bridge.

.8 The picture

Put into words, the two-layer architecture says:

Nature reckons in ratios.
We measure in SI.
The fractal geometry sits in the translator.

This is not only a pedagogical rule of thumb but a structural statement about FFGFT. It explains why some predictions are exact and others receive a 1.3% correction; it explains why K_{frak} is not a fitting parameter; it explains why the theory operates with one free parameter; and it makes precise the relation between geometric idealization and experimental measurement.

The epistemic point runs: *what we call "absolute values in SI" are human artifacts*. The universe "knows" nothing of meter, second, or kilogram. It knows only ratios, winding numbers, phases. The fractal corrections are the price we pay for measuring the universe *through our arbitrary units*. They are not a defect of the world but a property of the language in which we describe the world.

.9 The deeper point: the universe does not compute

Even the formulation "Layer 1 *computes* ratios, Layer 2 *computes* the SI values" is still too mechanistic. In neither layer does any computation take place. The universe does not compute; it *is* the configuration.

Layer 1 is a *geometry of admissible mode configurations*. A mode either exists or does not exist. It is not "computed"; it is the geometric configuration itself. The ratios stand where they stand because the winding recursion has converged at that point — not because anyone has computed it.

Layer 2, the bridge, is likewise not a computation but a *translation*. K_{frak} is not computed but is the only geometrically possible bridge function. The scale staircase itself does not "compute"; it is there.

Computation takes place only where an observer wants to derive Layer 2 from Layer 1 — the computation happens in the *observer*, not in the world.

The same logic makes photonic computing media (TFLN crystals, Ising machines, analog resonators, quantum dots; see Docs. 161, 167, 173) so efficient: the medium does not "compute", it lies in the solution. The geometry of the admissible phase configurations *is* the result. What in a digital machine runs as a sequential operation in time is in the photonic medium a *simultaneous* configuration in space: not algorithm but geometry. FFGFT makes the same claim about nature as a whole — the mode structure of the 4D-torus is not the program that nature executes, but nature itself.

.10 Where we really are: c as the experience of transience

But the architecture is still not complete. Layer 1 is the geometric pure form; Layer 2 is the human scale description. Both are *languages* about something in which we ourselves actually are. But we live neither in the pure geometry nor in the SI values. We live in a third position that does not coincide with either layer.

What we actually experience is *transience*. For us c is not one. c has a finite value because we experience the light path as duration, as a distance that takes time. Not because c "really" were $299\,792\,458$ m/s —

in pure geometry $c = 1$ —, but because we sit inside the configuration and experience it as *passing*. Were we Layer 1 itself, we would no longer be observers with a lifespan, but the geometry. The values set to one belong to the eye of the geometry. We see from inside — and exactly there they take finite values, and exactly there they are inserted, as soon as we measure.

This is the sense of the statement that the values set to one must be reinserted when projected measurement values are generated: not because the geometry has them, but because *we who measure* are in a position where they take finite values. c is the phenomenological trace of our inside position. Without an observer who ages, there would be no finite c.

.11 Closing point: the fractal trace of our experience

And with this the architecture, which at the beginning looked merely static, closes into a dynamic form. The recursion was never only in the ratios (Layer 1) or only in the bridge (Layer 2). It is also in us.

Every moment in which we age contains all previous moments in damped form. Our experience is not a linear stream but a *concert hall* (Doc. 159): every new sound carries the echo of all old ones. Human memory is not a storage but a recursive resonance in which every recurrence is a faint copy of the previous one.

If the universe *is* the configuration and we *experience* the configuration, then our experience is the *fractal trace of the geometry in which we lie*. The recursion is not an abstract feature of the theory — it is what we recognize ourselves in as observers. The same pattern that has come to rest in the ratios and that is still in motion in the scale bridge is also the pattern of our perception.

This is the arc this document draws: from the apparent multiplicity of natural constants over the two layers and their recursive relation to the position of the observer, who does not *know* the geometry but *is* it — and yet runs through it as transience. FFGFT does not describe a machinery that someone looks at from outside. It describes a geometry that carries itself, and an observer who lies inside this geometry and must therefore express it in two languages.

Whoever has seen this no longer sees c, $\hbar$, α, and K_{frak} as separate natural constants but as different traces of the same one geometry

— seen from different positions within the one configuration that we do not observe but in which we are.

.12 Reading order: where each layer is worked out

For systematic study, the two layers are treated in the following documents:

- **Layer 1 (ratios, geometric pure form):** Doc. 002 (Journey), Doc. 015 (systematics of natural units), Doc. 009 (origin of ξ), Doc. 205 (narrative), Doc. 006 (particle masses as ratios), Doc. 060/116/159 (mode and winding ratios), Doc. 230 (Hilbert-space bridge, likewise ratio-based).
- **Layer 2 (scale bridge, fractal corrections):** Doc. 014 (transition nat./SI with scale factor S_{T0}), Doc. 013 (SI conventions), Doc. 133 (complete derivation of K_{frak}), Doc. 134 (c as convention), Doc. 044 (conventions for α), Doc. 185 (embedding cost of mathematics).

Whoever reads the documents in this order sees the architecture this document speaks of at concrete points: ratios above, bridge below, no free parameter except ξ_0.

Summary

- FFGFT operates on two cleanly separated layers: ratios (Layer 1, fractal-free) and SI projection (Layer 2, with fractal bridge).
- The values $c, \hbar, G, k_B, \alpha$ that have been "set to one" do not disappear but are factored out; they return when projected measurement values are generated.
- The recursion is present in the ratios as recursion at rest (fixed points of a winding sequence with Fibonacci convergence at $n \approx 8$) and in the bridge as recursion in motion ($K_{\text{frak}} = 1 - 100\xi$).
- The universe does not compute; it is the configuration. Computation happens in the observer — analogous to the logic of photonic computing media (TFLN, Ising machines).

- We live neither in Layer 1 nor in Layer 2 but in a third position: the experience of transience. c has for us a finite value because we *traverse* the configuration instead of *being* it.
- The recursion also shows itself in our experience: every moment carries the echo of all previous ones. Our perception is the fractal trace of the geometry in which we lie.

Status: May 2026. This document is methodologically pedagogical and contains no independent physical claims; all factual statements refer to the source documents cited.